

CHAPTER 10

Muscle Tissue

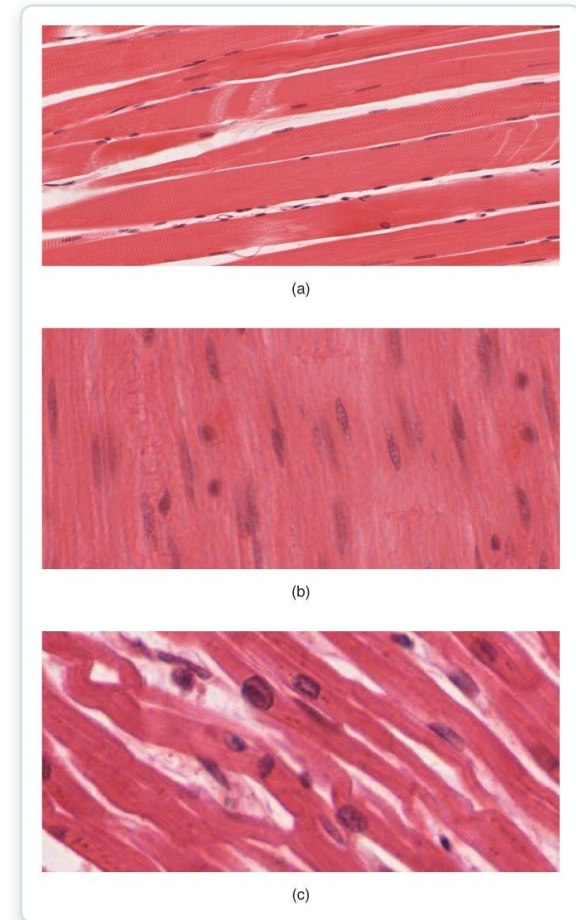
How muscle contracts, from sarcomere to whole-body movement.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Explain the organization of muscle tissue
- 2 Describe the function and structure of skeletal, cardiac muscle, and smooth muscle
- 3 Explain how muscles work with tendons to move the body
- 4 Describe how muscles contract and relax
- 5 Define the process of muscle metabolism
- 6 Explain how the nervous system controls muscle tension



The three types of muscle tissue.



10.1

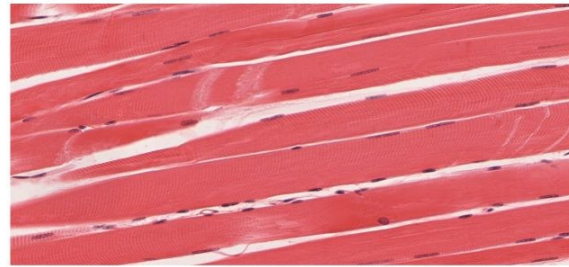
SECTION

Overview of Muscle Tissues

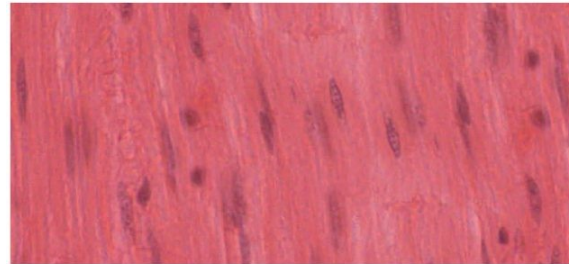
The three muscle types and the shared properties of muscle tissue.

SECTION 10.1

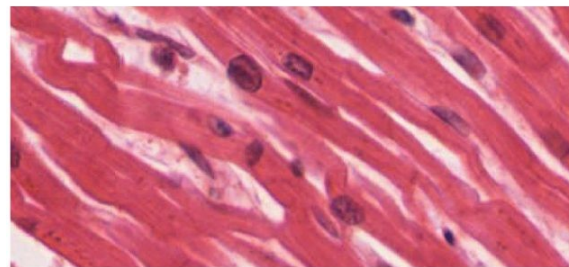
The Three Muscle Types



(a)



(b)



(c)

Skeletal, cardiac, and smooth muscle

- Skeletal, cardiac, and smooth
- All are excitable and contractile
- Skeletal is voluntary and striated
- Cardiac and smooth are involuntary



10.2

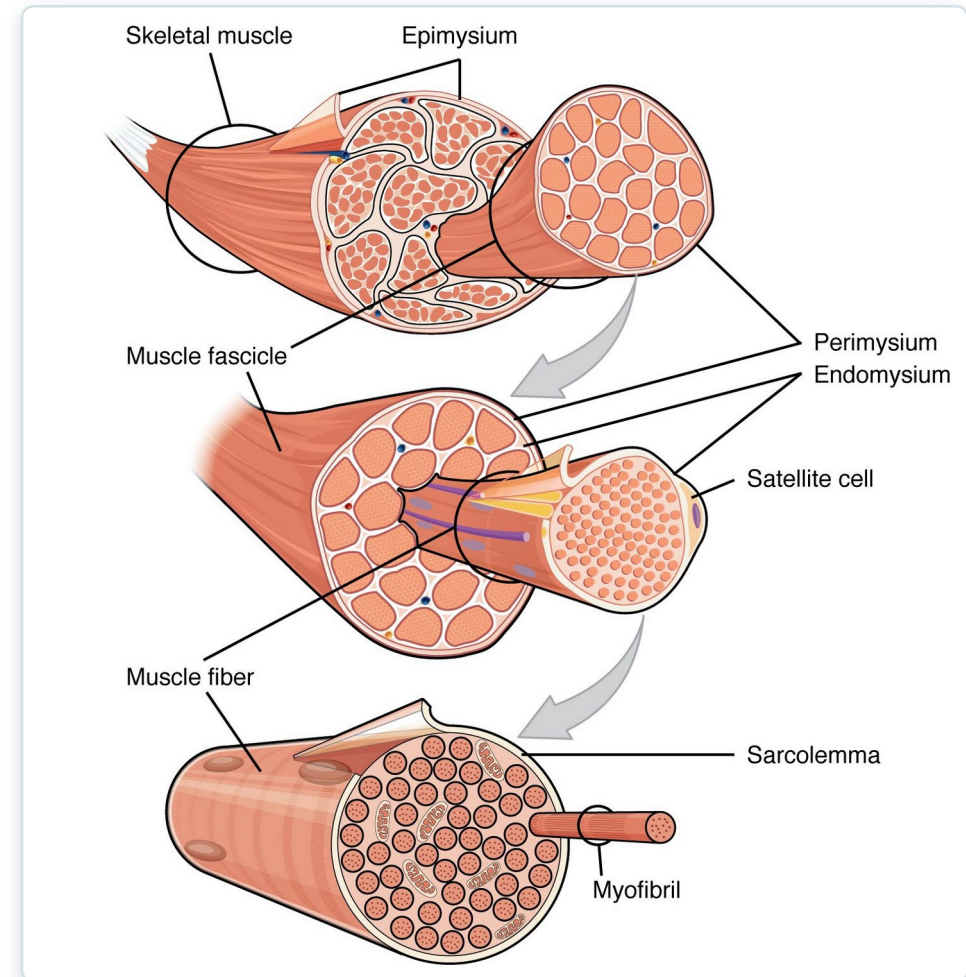
SECTION

Skeletal Muscle

The organized structure of skeletal muscle, from whole muscle to sarcomere.

Organization of Skeletal Muscle

- Muscle → fascicles → fibers → myofibrils
- Connective sheaths wrap each level
- Epimysium, perimysium, endomysium
- Sheaths merge into the tendon

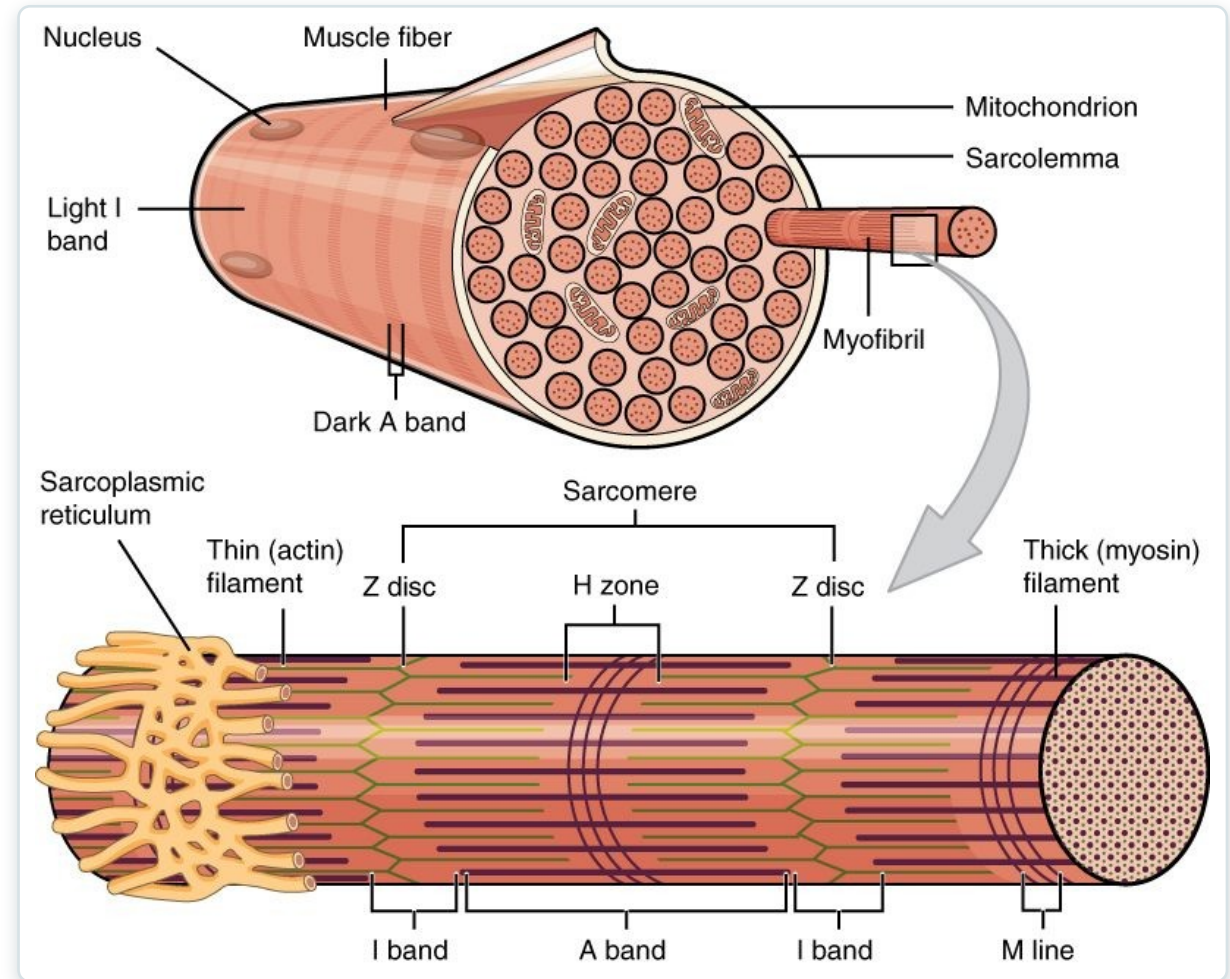


The connective-tissue organization of skeletal muscle

SECTION 10.2

The Muscle Fiber

- A muscle cell is called a fiber
- Sarcolemma is its membrane
- Myofibrils run its length
- Striations come from sarcomeres

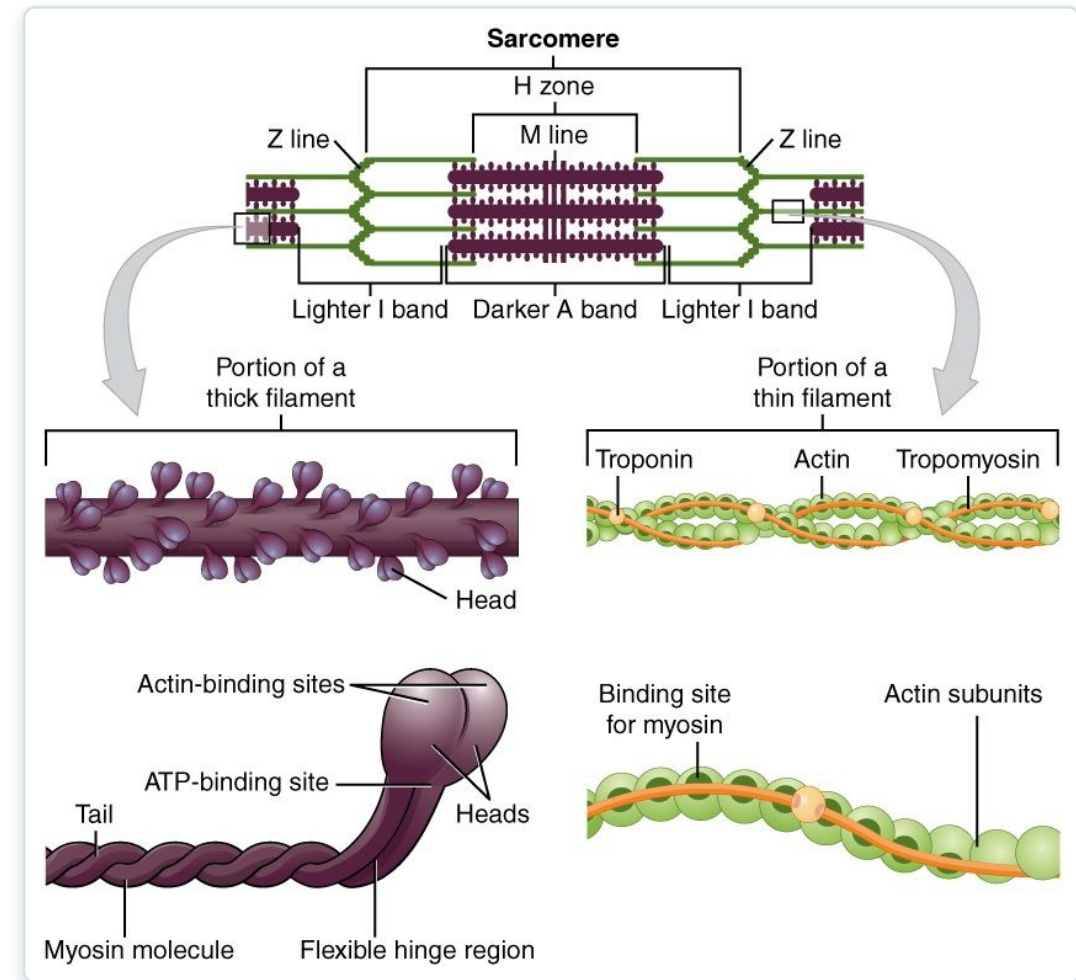


Structure of a skeletal muscle fiber

SECTION 10.2

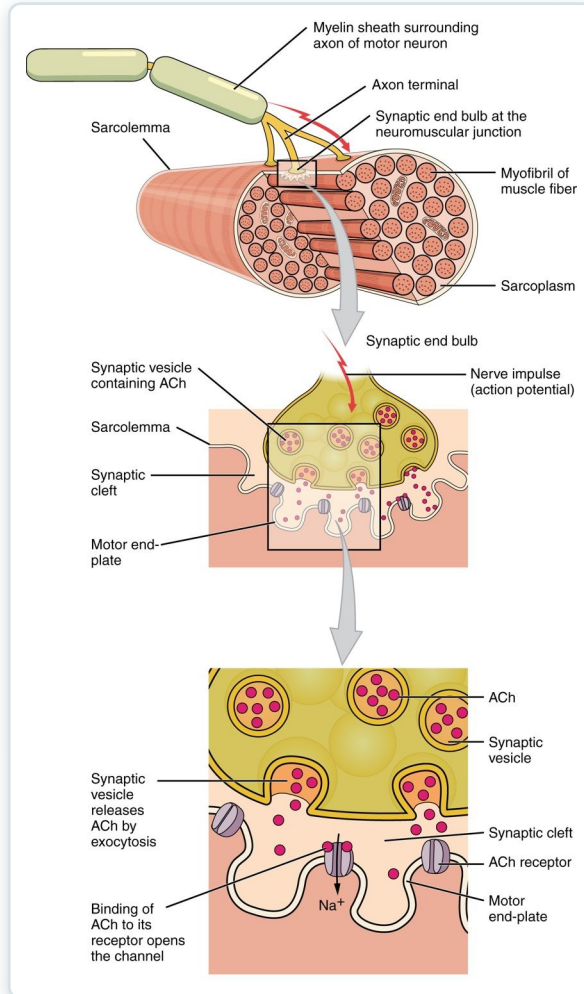
The Sarcomere

- The contractile unit of muscle
- Thick (myosin) and thin (actin) filaments
- Bounded by Z lines
- Troponin and tropomyosin regulate it



The sarcomere and its filaments

The Neuromuscular Junction



The neuromuscular junction

- Where a motor neuron meets a fiber
- Releases acetylcholine (ACh)
- ACh triggers an action potential
- Begins excitation-contraction coupling

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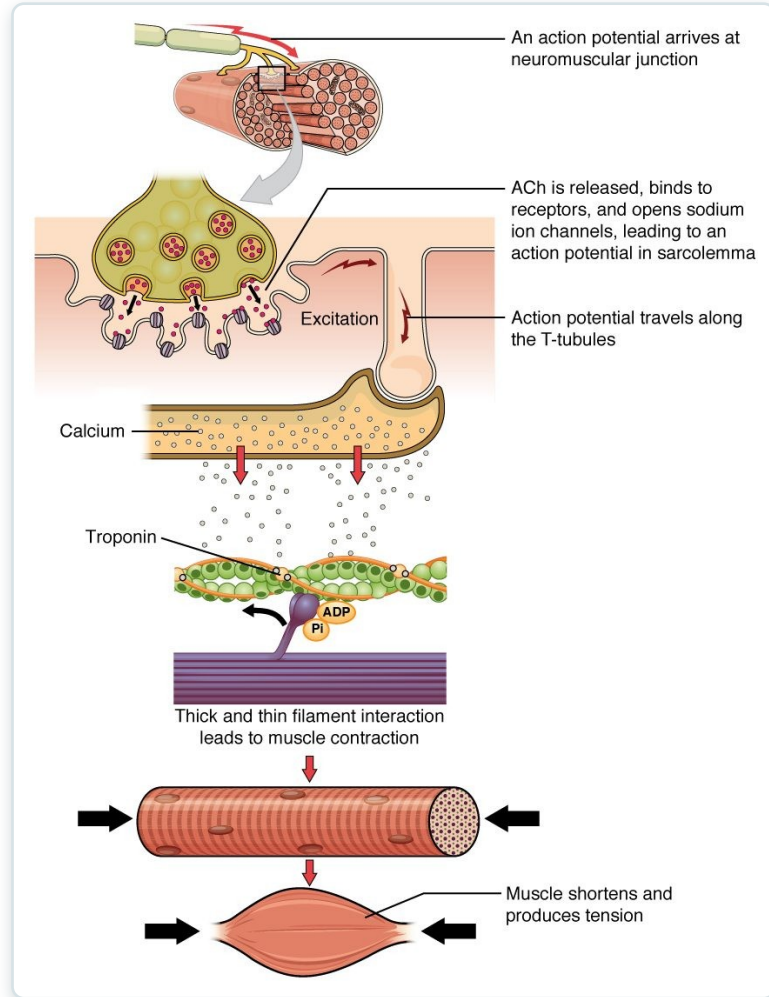
10.3

SECTION

Muscle Fiber Contraction and Relaxation

How a muscle fiber contracts and relaxes — the sliding filament model.

Excitation-Contraction Coupling



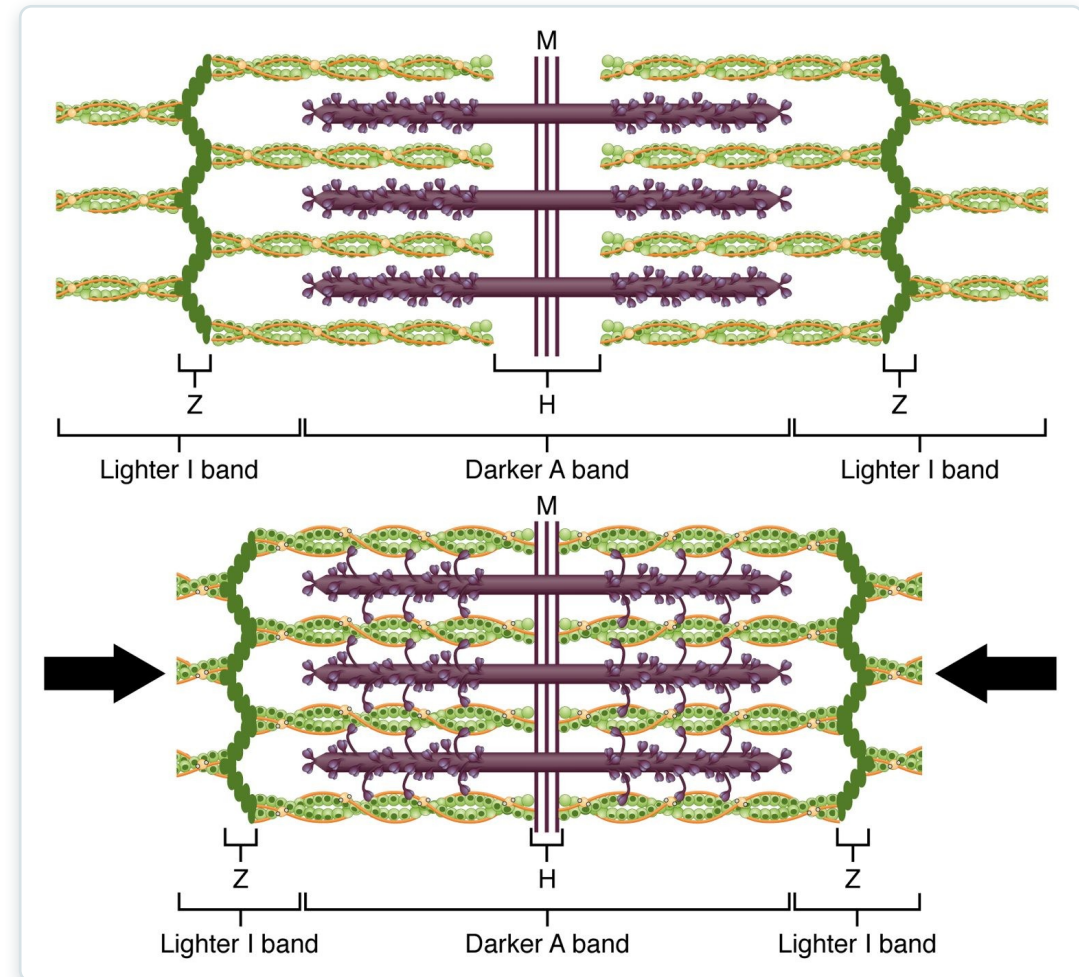
The steps of muscle contraction

- Action potential reaches the fiber
- Calcium is released from the SR
- Calcium exposes actin binding sites
- Myosin pulls actin → contraction

SECTION 10.3

The Sliding Filament Model

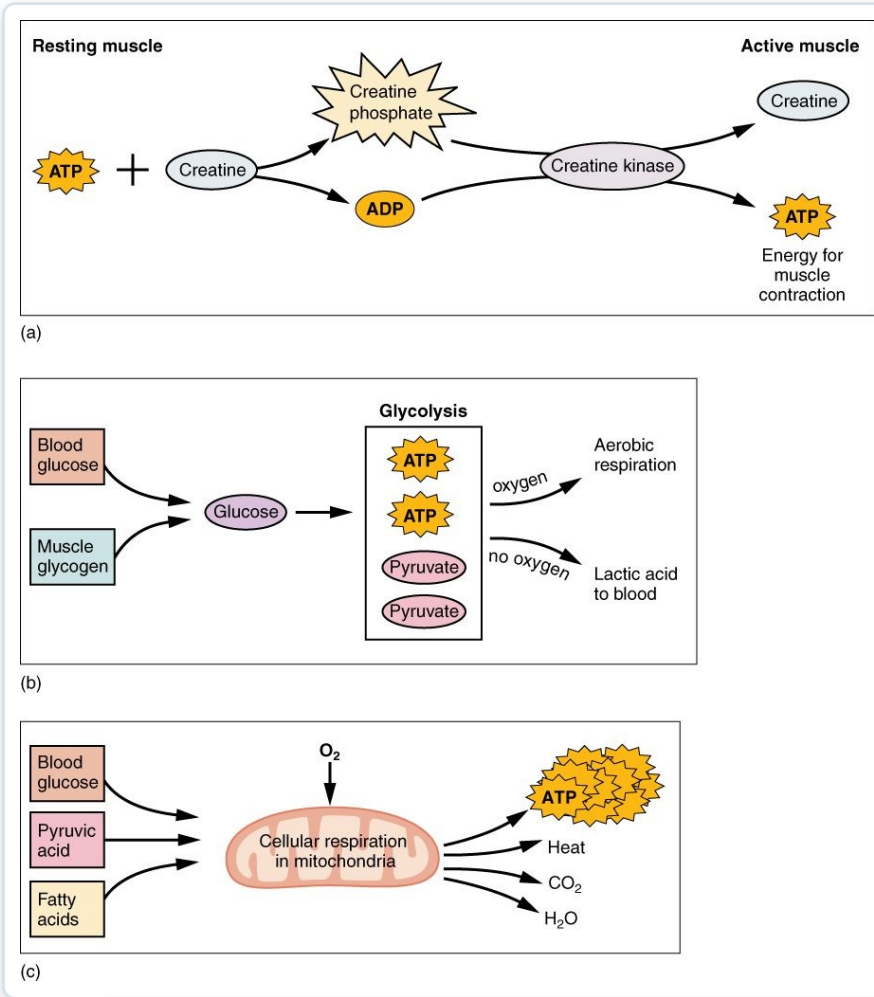
- Filaments slide; they don't shorten
- Myosin heads pull actin inward
- The sarcomere shortens
- The whole muscle contracts



The sliding filament model of contraction

SECTION 10.3

Energy for Contraction



Sources of ATP for muscle contraction

- Muscles run on ATP
- Creatine phosphate gives a quick burst
- Aerobic respiration for endurance
- Anaerobic glycolysis makes lactic acid



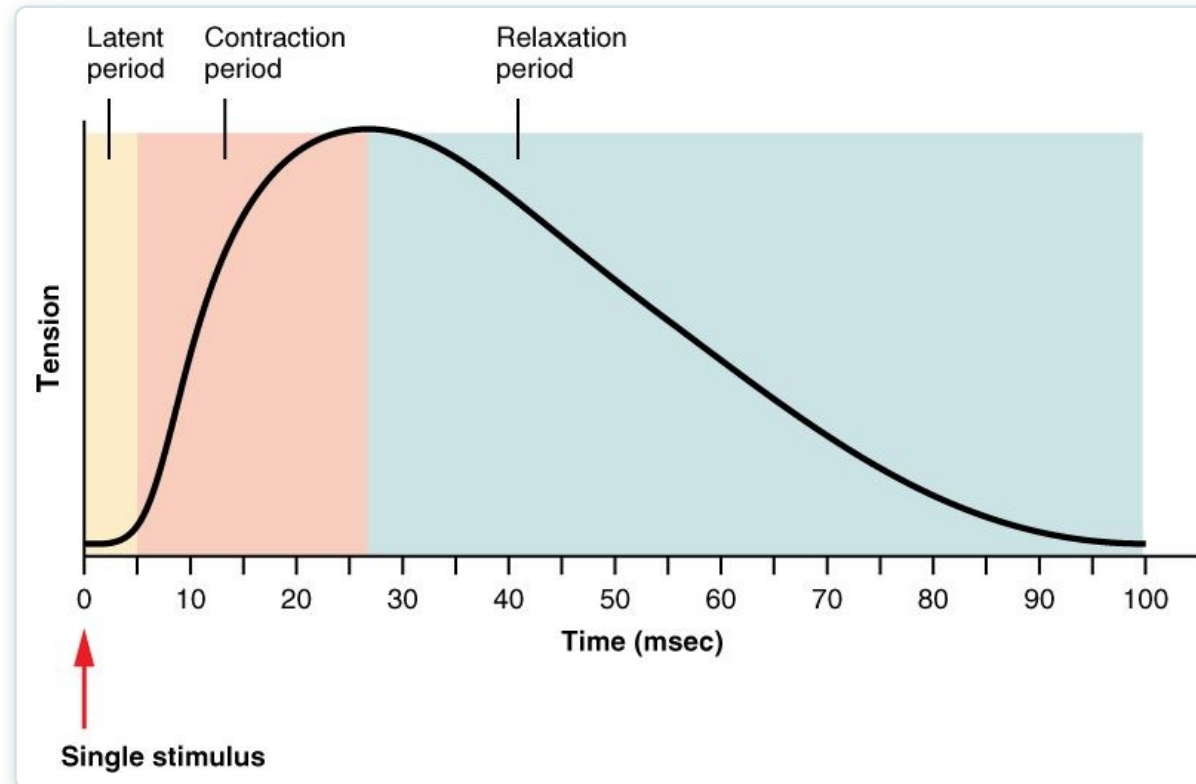
10.4

SECTION

Nervous System Control of Muscle Tension

How the nervous system grades the strength of contraction.

The Muscle Twitch



The phases of a muscle twitch

- A twitch is one contraction–relaxation cycle
- Latent, contraction, and relaxation periods
- Motor units grade fine vs. gross movement
- Summation increases force



10.5

SECTION

Types of Muscle Fibers

Slow and fast muscle fibers suited to endurance or power.

Types of Muscle Fibers

- Slow-twitch fibers resist fatigue
- Suited to endurance activities
- Fast-twitch fibers contract powerfully
- They fatigue more quickly



Endurance depends on muscle fiber type



10.6

SECTION

Exercise and Muscle Performance

How exercise and disuse change muscle size and performance.

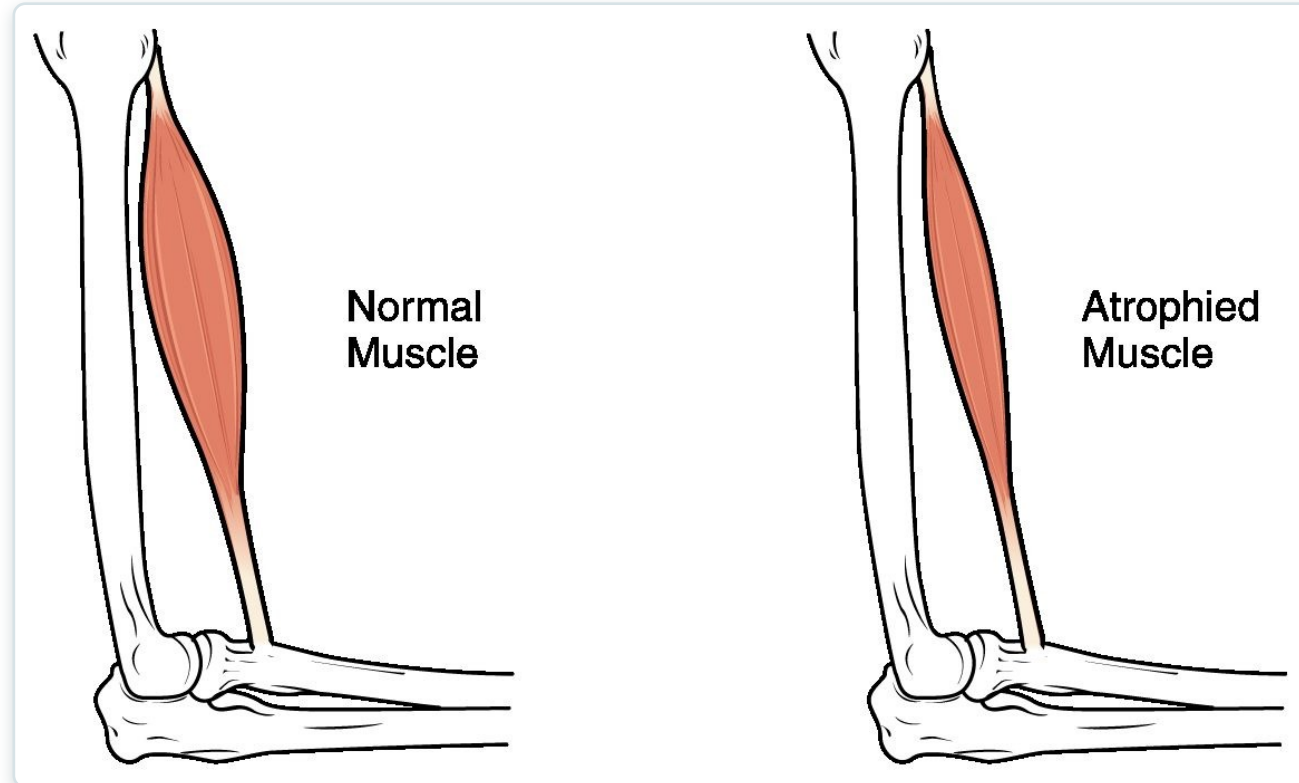
Exercise & Hypertrophy



Muscle hypertrophy from exercise

- Resistance exercise enlarges fibers
- Builds strength and power
- Endurance exercise boosts stamina
- Improves blood supply and metabolism

Muscle Atrophy



Normal versus atrophied muscle

- Disuse shrinks muscle (atrophy)
- Occurs with immobility or bed rest
- Strength and mass decline quickly
- Prevented by movement and exercise



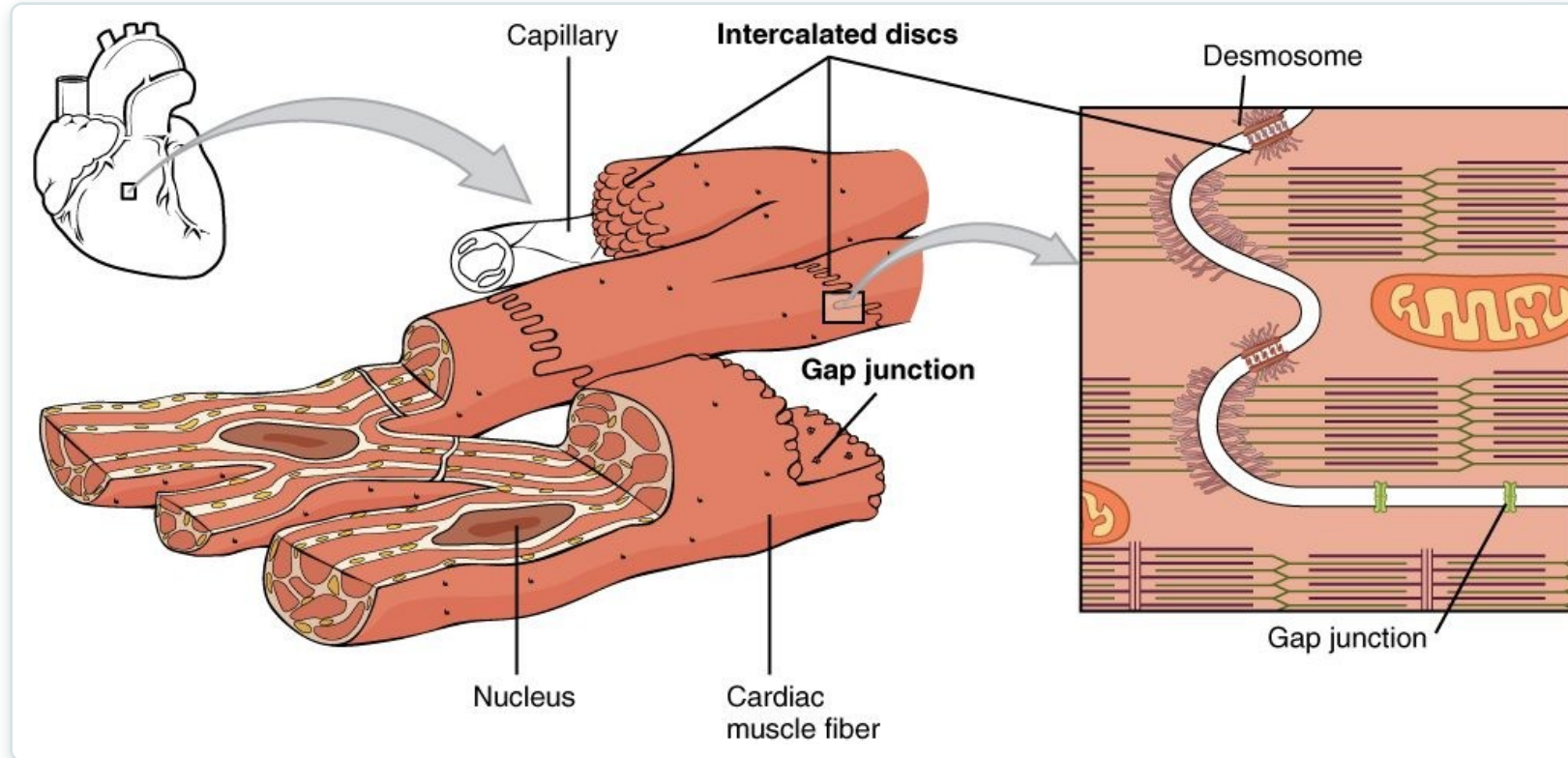
10.7

SECTION

Cardiac Muscle Tissue

The involuntary striated muscle that pumps the heart.

Cardiac Muscle



Cardiac muscle tissue

- Found only in the heart
- Striated but involuntary
- Cells joined by intercalated discs
- Gap junctions coordinate the beat



10.8

SECTION

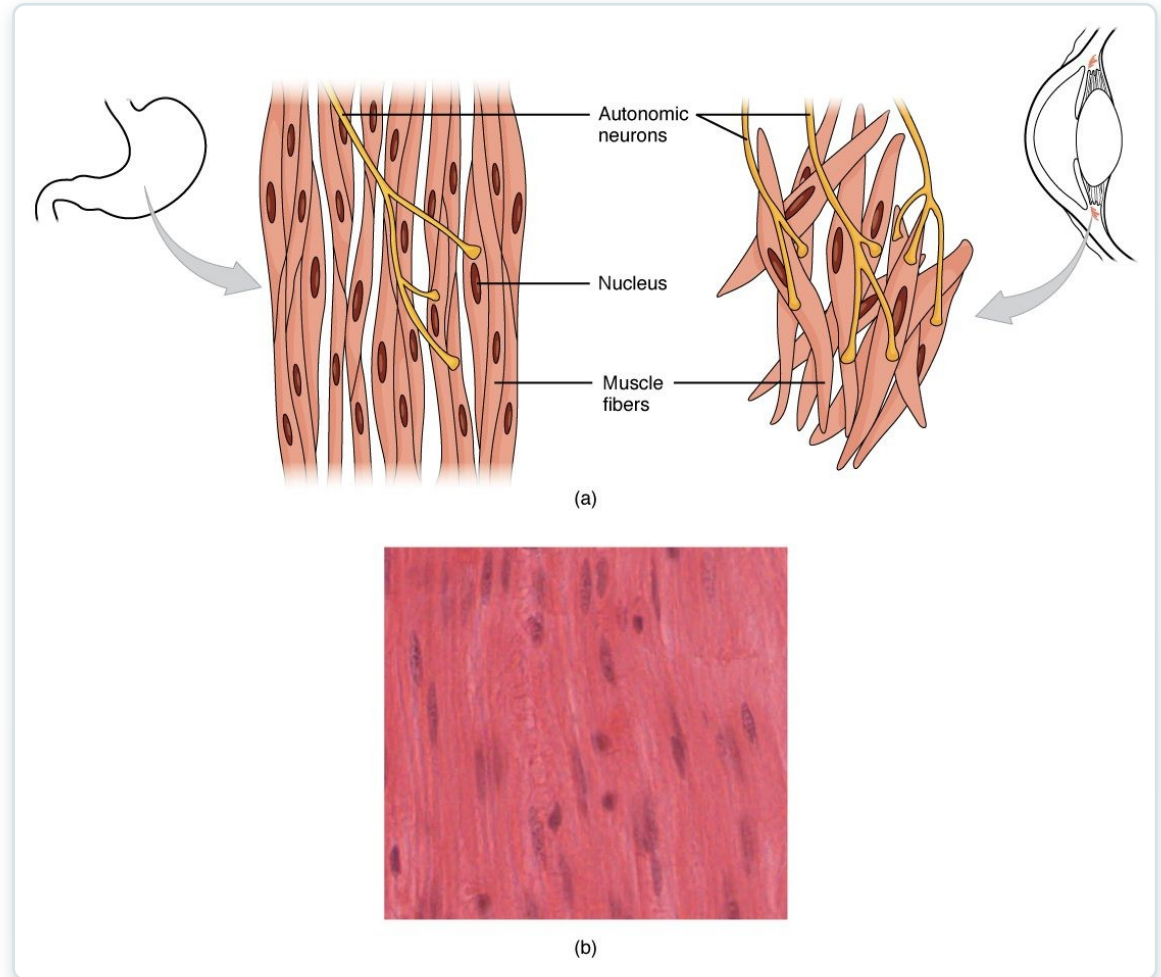
Smooth Muscle

The involuntary muscle of hollow organs and vessels.

SECTION 10.8

Smooth Muscle

- Lines hollow organs and vessels
- Involuntary and non-striated
- Slow, sustained contractions
- Moves food, blood, and urine



Smooth muscle tissue

Key Takeaways

- The three muscle types — skeletal, cardiac, and smooth — share excitability and contractility.
- Skeletal muscle is organized from the whole muscle down to the sarcomere.
- Contraction follows the sliding filament model and requires calcium and ATP.
- The nervous system grades contraction strength through motor-unit recruitment.
- Cardiac and smooth muscle are involuntary, specialized for the heart and hollow organs.

CLINICAL CONNECTIONS

Why Muscle Tissue Matters in Practical Nursing

Mobility & Atrophy

Disuse weakens muscle fast — early movement prevents decline.

Neuromuscular Drugs

Many medications act at the neuromuscular junction.

Energy & Fatigue

ATP supply and lactic acid shape exercise tolerance.

Cardiac Muscle

Coordinated contraction underlies every heartbeat and EKG.

Smooth Muscle

It controls blood pressure, airflow, and GI motility.

Rehabilitation

Understanding contraction guides strengthening and recovery.